

Delay-Doppler Domain Channel Estimations for OTFS Systems with Time-Domain Pilots

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Abstract—A new channel estimation algorithm for orthogonal time-frequency space (OTFS) with time-domain pilots is proposed in this paper. Motivated by the fact that rapidly changing channel coefficients in the time domain are non-linear functions of slow changing propagation parameters in the DD domain, we propose to estimate the DD-domain propagation parameters by using time-domain pilots. This is different from DD-domain pilots with two-dimensional (2D) guard intervals adopted by most existing OTFS channel estimation methods. The employment of time-domain pilots only requires one-dimensional (1D) guard interval to counter interference introduced by delay spread. Specifically, we propose to estimate the Doppler shifts by exploiting the subspace structure of time-domain pilots with the multiple signal classification (MUSIC) algorithm. A new pilot decimation method is proposed to estimate the pilot covariance matrix necessary for the MUSIC algorithm. The MUSIC-based channel estimation algorithm allows us to decouple the highly non-linear relationship among the various DD-domain channel parameters, including Doppler shifts, delays and fading coefficients, such that the channel state information (CSI) can be estimated by considering the effects of propagation parameters and the ambiguity function of practical pulse shaping filters. Simulation results demonstrate that the proposed channel estimation algorithm with a small amount of time-domain pilot overhead can achieve a performance close to that of systems with perfect CSI.

I. INTRODUCTION

Delay-Doppler (DD) domain communications, such as orthogonal time frequency space (OTFS) [1] and orthogonal delay Doppler modulation (ODDM) [2], are promising modulation schemes for high mobility communication systems experiencing time and frequency doubly-selective fading. DD-domain waveforms are inherently resilient to fast time-varying channels, as channel coefficients in the DD domain change at a much slower rate compared to their time-frequency (TF) domain counterparts.

Channel estimation is essential for OTFS system developments, given that channel state information (CSI) is needed for all practical OTFS receivers [3], [4]. Almost all existing OTFS channel estimation methods are performed by inserting pilots in the DD domain [5]–[9]. The delay and Doppler shifts of doubly-selective fading will spread each data or pilot symbol across multiple DD-domain grid points and cause strong DD-domain inter-symbol interference (ISI). The situation is exacerbated with delay and Doppler shifts that are not integer multiples of the DD domain grid resolution [10]. In [5] and [6], OTFS channel estimations are performed by using entire frames dedicated for pilot transmissions to

avoid ISI between pilot and data symbols, and this leads to low spectral efficiency due to large pilot overhead. A more commonly used approach is to place two-dimensional (2D) guard intervals between pilot and data symbols [7]. To further reduce pilot overhead, pilots are superimposed on data symbols and rough channel estimations are performed by treating data symbols as interference, then iterative data-aided estimation-detection is performed to refine the estimation accuracy [8], [9]. All of the above works rely on separate processing of integer and fractional Doppler spreads, which results in large overhead with high complexity.

Instead of estimating the DD-domain channel transfer function (CTF) as in [7], [8], several methods have been developed to estimate the DD-domain channel propagation parameters such as delay, Doppler, and fading coefficients, by using subspace-based algorithms such as multiple signal classification (MUSIC) [11], [12] or matched filtering with a pseudo-noise (PN) sequence [6]. All these works still rely on DD-domain pilot patterns. A TF pilot pattern is adopted in [13] to reduce peak-to-average power ratio, and it still requires a 2D guard interval with relatively high overhead.

In this paper, we propose a new OTFS channel estimation method by using time-domain pilot symbols. Inserting pilot symbols in the time domain requires one-dimensional (1D) guard intervals to counter the delay spread, and this is different from the 2D guard intervals necessary for DD-domain pilot patterns. In the proposed approach, pilot patterns are inserted among the time domain data samples at the transmitter and removed before converting the time domain samples back to the DD domain, thus the time-domain pilots do not interfere with the DD-domain data symbols.

Unlike existing methods that either estimate the equivalent DD-domain CTF or propagation parameters, the proposed channel estimation algorithm is developed by using a hybrid approach that reconstructs the time- or DD-domain CTF by using partial knowledge of the DD domain propagation parameters. Specifically, we propose to estimate the Doppler shifts by applying the MUSIC algorithm to the time domain pilots, because Doppler shifts are manifested as weighted sums of complex sinusoids in the time domain. The unknown coefficients of the complex sinusoids are non-linear functions of fading coefficients and the delay-Doppler shifted ambiguity functions of the prototype pulse, and these coefficients are estimated by exploiting the DD-domain structure of the time-

domain pilot symbols. A new pilot decimation method is proposed to estimate the pilot covariance matrix required by the MUSIC algorithm. The newly proposed channel estimation algorithm allows us to reconstruct the time- or DD-domain CTF by decoupling the highly non-linear relationship among the various propagation parameters and the ambiguity function of the prototype pulses, and it can be directly applied to channels with arbitrary delay and Doppler shifts without the need to separately consider integer and fractional Doppler shifts. Simulation results show that the proposed channel estimation algorithm with small pilot overheads can achieve a performance that is close to that of systems with perfect CSI.

II. SYSTEM MODEL

The OTFS system model with the proposed pilot scheme is presented in this section. In the TF domain, each OTFS frame contains N time symbols with symbol interval T , and M subcarriers with subcarrier spacing $\Delta f = \frac{1}{T}$. The DD domain is discretized into an $M \times N$ grid with M and N levels in the delay and Doppler domains, respectively. The minimum delay domain interval is $\Delta\tau = \frac{1}{M\Delta f}$, and the minimum Doppler spacing is $\Delta\nu = \frac{1}{NT}$. Denote $x^{\text{DD}}[l, k]$ as the information symbol transmitted at delay $l\Delta\tau$ and Doppler shift $k\Delta\nu$, for $l = 0, \dots, M-1$ and $k = 0, \dots, N-1$.

At the transmitter, the DD-domain symbols are converted to the TF domain by using the inverse Symplectic finite Fourier transform (ISFFT) as

$$x^{\text{TF}}[m, n] = \frac{1}{MN} \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} x^{\text{DD}}[l, k] e^{j2\pi(\frac{nk}{N} - \frac{ml}{M})}, \quad (1)$$

where $x^{\text{TF}}[m, n]$ is the TF sample to be transmitted at subcarrier $m\Delta f$ and time nT . The discrete TF-domain samples are converted to discrete time-domain samples by performing M -point inverse discrete Fourier transform (IDFT) as

$$s[k, n] = \frac{1}{\sqrt{M}} \sum_{m=0}^{M-1} x^{\text{TF}}[m, n] e^{j2\pi \frac{mk}{M}}, \quad (2)$$

for $k = 0, 1, \dots, M-1, n = 0, 1, \dots, N-1$.

A. Pilot Patterns

Pilot patterns are inserted among the time-domain samples $s[k, n]$ at the transmitter, and then removed at the receiver before converting the time-domain samples back to the DD domain. Thus, the pilot patterns will not affect the DD-domain information symbols.

Define the pilot pattern as $\mathbf{p} = [\mathbf{0}_L^T, s_p, \mathbf{0}_L^T]^T \in \mathcal{C}^{(2L+1) \times 1}$, where s_p is a pilot symbol with energy $\mathbb{E}[|s_p|^2] = E_p$, and $\mathbf{0}_L$ is a length- L all-zero vector, with $L+1$ being the length of the delay-domain support of the equivalent discrete-time channel transfer function, which will be defined in the next subsection.

Assume U equally-spaced pilot patterns are inserted in each OTFS frame as shown in Fig. 1, where $s_{k+nM} = s[k, n]$. The number of pilot patterns U is chosen as a factor of NM , and

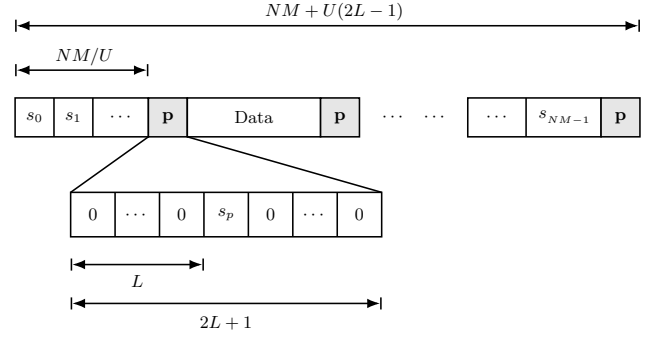


Fig. 1: Pilot pattern scheme

U needs to be larger than the number of propagation paths of the physical doubly-selective fading channel. Denote the time-domain vector containing both information samples and pilot patterns as shown in Fig. 1 as $\mathbf{a} = [a[0], a[1], \dots, a[B-1]]^T \in \mathcal{C}^B$, with $B = NM + (2L+1)U$.

B. Equivalent Discrete-Time Model

The discrete time samples have to be pulse shaped before transmission. In this paper, we consider an OTFS system with sample-wise prototype pulse as proposed in [4]. Denote the transmission pulse employed by the OTFS system as $q_{\text{TX}}(t)$ with bandwidth proportional to $\frac{1}{T_s} = \frac{M}{T}$; then the continuous-time transmitted signal can be represented as

$$s(t) = \sum_{n=0}^{B-1} a[n] q_{\text{TX}}(t - nT_s), \quad (3)$$

where $T_s = \frac{T}{M}$. The continuous-time signal is transmitted through a doubly-selective channel with DD-domain channel impulse response as

$$\varphi(\tau, \nu) = \sum_{i=1}^P \varphi_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (4)$$

where $\delta(\cdot)$ is the Dirac delta function, φ_i , τ_i , and ν_i are the path gain, delay, and Doppler shift for the i -th path, respectively, and P is the number of propagation paths. The signal observed at the receiver is

$$r(t) = \sum_{i=1}^P s(t - \tau_i) e^{j2\pi\nu_i(t - \tau_i)} + z(t), \quad (5)$$

where $z(t)$ is additive white Gaussian noise (AWGN) with one-sided power spectral density N_0 . The received signal passes through a matched filter $q_{\text{RX}}(t) = q_{\text{TX}}^*(-t)$. The signal at the output of the matched filter, $\eta(t) = r(t) \otimes q_{\text{TX}}^*(-t)$, can be represented as

$$\begin{aligned} \eta(t) &= \sum_{i=1}^P \varphi_i \int q_{\text{TX}}^*(t' - t) s(t' - \tau_i) e^{j2\pi\nu_i(t' - \tau_i)} dt' + w(t), \\ &= \sum_{n=0}^{B-1} \sum_{i=1}^P a[n] \varphi_i e^{j2\pi\nu_i nT_s} A_q(t - \tau_i - nT_s, \nu_i) + w(t), \end{aligned} \quad (6)$$

where $w(t) = \int_{-\infty}^{\infty} q_{\text{TX}}^*(t' - t)z(t')dt'$ is the noise component(s?) at the output of the matched filter, and $A_q(t, f)$ is the ambiguity function of $q_{\text{TX}}(t)$ defined as

$$A_q(t, f) = \int_{-\infty}^{\infty} q_{\text{TX}}^*(\tau - t)q_{\text{TX}}(\tau)e^{j2\pi f\tau}d\tau. \quad (7)$$

The output of the matched filter is sampled at a sampling period T_s . The time domain samples can be represented as

$$\eta[k] = \sum_{n=0}^{B-1} a[n]h_k[k - n] + w[k], \quad (8)$$

where $h_k[l]$ is the equivalent discrete-time channel transfer function defined as

$$h_k[l] = \sum_{i=1}^P \varphi_i e^{j2\pi\nu_i(k-l)T_s} A_q(lT_s - \tau_i, \nu_i). \quad (9)$$

We consider a practical rectangular waveform as $q_{\text{TX}}(t)$ with a duration T_s seconds, and the closed-form expression of the ambiguity function $A_q(t)$ can be found in [14]. Due to the finite support of the waveform in the time domain, $h_k[l]$ has finite support on $l \in [0, 1 + \lfloor \frac{\tau_{\max}}{T_s} \rfloor]$, where $\tau_{\max} = \max(\tau_i)$ is the maximum delay spread. Thus we have

$$L = 1 + \lfloor \frac{\tau_{\max}}{T_s} \rfloor. \quad (10)$$

After removal of CP at the receiver, the time-domain samples, $\eta[k]$, can be represented in a matrix format as

$$\boldsymbol{\eta} = \mathbf{H}\mathbf{a} + \mathbf{w}, \quad (11)$$

where $\mathbf{H} \in \mathbb{C}^{B \times B}$ is defined in (12), and $\boldsymbol{\eta}$ and $\mathbf{w}$ are size- B vectors with elements being $\eta[k]$ and $w[k]$, respectively.

III. CHANNEL ESTIMATION

A. Pilot Extraction

We first extract the received time domain samples that are contributed exclusively by the pilot symbols. The pilot patterns divide the received samples into U blocks, each of length $\frac{B}{U} = \frac{NM}{U} + 2L + 1$. In each block, the first $\frac{NM}{U} + L$ samples are contributed by the information samples, and the last $L + 1$ samples are contributed exclusively by the pilot symbol s_p . Thus we can reorganize the received sample vector $\boldsymbol{\eta}$ into a $\frac{B}{U} \times U$ matrix $\boldsymbol{\Psi}$, which is constructed by filling the matrix column-wise with the elements of $\boldsymbol{\eta}$ as

$$\boldsymbol{\Psi} = \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \cdots & \mathbf{y}_U \\ \mathbf{z}_1 & \mathbf{z}_2 & \cdots & \mathbf{z}_U \end{bmatrix} \in \mathbb{C}^{\frac{B}{U} \times U}, \quad (13)$$

where $\mathbf{y}_u \in \mathbb{C}^{\frac{NM}{U} + L}$ and $\mathbf{z}_u \in \mathbb{C}^{L+1}$ (should we make $\mathbf{y}_u$ and $\mathbf{z}_u$ more explicit) are the received information and pilot sample vectors of the u -th block, respectively.

The channel estimation will be performed by using the pilot sample vectors $\mathbf{z}_u$, for $u = 1, \dots, U$. To simplify notation, define $l_u = (u - 1)\frac{B}{U} + \frac{NM}{U} + L + 1$ as the index of the first received pilot sample in the u -th block. With the length- L all-zero guard interval following each pilot symbol, the l -th element of $\mathbf{z}_u$ can be expressed as

$$z_u[l] = s_p \cdot h_{l_u+l}[l] + w[l_u + l], \text{ for } l = 0, \dots, L. \quad (14)$$

B. Doppler Estimation

The Doppler shifts, $\{\nu_i\}_{i=1}^P$, will be estimated by using the received pilot samples with the MUSIC algorithm.

From (9), the equivalent discrete-time channel coefficient $h_{k+l}[l]$ can be alternatively expressed as

$$h_{k+l}[l] = [e^{j\omega_1 k}, e^{j\omega_2 k}, \dots, e^{j\omega_P k}] \cdot \boldsymbol{\beta}_l, \quad (15)$$

where $\omega_i = 2\pi\nu_i T_s$ is the normalized angular Doppler shift, and

$$\boldsymbol{\beta}_l = [\varphi_1 A_q(lT_s - \tau_1, \nu_1), \dots, \varphi_P A_q(lT_s - \tau_P, \nu_P)]^T, \quad (16)$$

is a length- P vector containing the unknown fading coefficients φ_i and the delay-Doppler-shifted ambiguity function $A_q(lT_s - \tau_i, \nu_i)$.

For a given l , stacking $\{z_u[l]\}_{u=1}^U$ into a column vector as $\boldsymbol{\zeta}_l = [z_1[l], z_2[l], \dots, z_U[l]]^T \in \mathbb{C}^U$. From (14) and (15), we have

$$\boldsymbol{\zeta}_l = s_p \cdot \mathbf{E}_U \cdot \boldsymbol{\beta}_l + \mathbf{w}_l, \quad (17)$$

where $\mathbf{w}_l = [w(l_1 + l), \dots, w(l_U + l)]^T \in \mathbb{C}^U$ is the noise vector, and $\mathbf{E}_U$ is a Vandermonde matrix defined as

$$\mathbf{E}_U = \begin{bmatrix} e^{j\omega_1 l_1} & e^{j\omega_2 l_1} & \cdots & e^{j\omega_P l_1} \\ e^{j\omega_1 l_2} & e^{j\omega_2 l_2} & \cdots & e^{j\omega_P l_2} \\ \vdots & \vdots & \ddots & \vdots \\ e^{j\omega_1 l_U} & e^{j\omega_2 l_U} & \cdots & e^{j\omega_P l_U} \end{bmatrix} \in \mathbb{C}^{U \times P}. \quad (18)$$

The pilot sample vector $\boldsymbol{\zeta}_l$ takes U equally-spaced samples from the l -th diagonal of the time-domain channel matrix $\mathbf{H}$, with the space between two adjacent samples being $l_{u+1} - l_u = \frac{B}{U}$ samples. Both $\mathbf{E}_U$ and $\boldsymbol{\beta}_l$ in (17) contain unknown information. The Vandermonde matrix $\mathbf{E}_U$ depends on the Doppler shifts $\{\nu_i\}_{i=1}^P$, and the vector $\boldsymbol{\beta}_l$ depends on all unknown channel parameters $\{\varphi_i, \nu_i, \tau_i\}_{i=1}^P$.

We propose to estimate the Doppler shifts by exploiting the unique structure of $\mathbf{E}_U$, which allows us to obtain estimates of Doppler shifts by using $\boldsymbol{\zeta}_l$ without knowing $\boldsymbol{\beta}_l$. Given $\{\nu_i, \tau_i\}_{i=1}^P$, both $\boldsymbol{\beta}_l$ and $\mathbf{w}_l$ are zero-mean Gaussian vectors, thus $\boldsymbol{\zeta}_l$ is zero-mean Gaussian distributed with covariance matrix

$$\mathbf{R}_l = \mathbb{E}[\boldsymbol{\zeta}_l \boldsymbol{\zeta}_l^H] = E_p \mathbf{E}_U \mathbf{R}_{\boldsymbol{\beta}_l} \mathbf{E}_U^H + \sigma_0^2 \mathbf{I}_U, \quad (19)$$

where $\mathbf{I}_U$ is a size- U identity matrix, σ_0^2 is the noise variance, and $\mathbf{R}_{\boldsymbol{\beta}_l}$ is a $P \times P$ diagonal matrix with the p -th diagonal element being $\sigma_p^2 \cdot |A_q(lT_s - \tau_p, \nu_p)|^2$, with $\sigma_p^2 = \mathbb{E}(|\varphi_p|^2)$. The covariance matrix $\mathbf{R}_l$ can be estimated by using a sample average with a new pilot decimation method, and details will be discussed in the next subsection.

Performing eigenvalue decomposition of $\mathbf{R}_l$ and sorting the eigenvalues in descending order, we can divide the eigenspace of $\mathbf{R}_l$ into the signal and noise subspaces. The first P eigenvectors of $\mathbf{R}_l$ span the signal subspace given that $\mathbf{E}_U$ has rank U , and the remaining $U - P$ eigenvectors span the noise subspace, and the two subspaces are orthogonal to each other.

$$\mathbf{H} = \begin{bmatrix} h_0[0] & 0 & 0 & \cdots & 0 & h_0[L] & \cdots & \cdots & h_0[1] \\ h_1[1] & h_1[0] & 0 & \cdots & 0 & 0 & h_1[L] & \cdots & h_1[2] \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ h_L[L] & \cdots & h_L[0] & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ 0 & h_{L+1}[L] & \cdots & h_{L+1}[0] & 0 & \cdots & \cdots & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 & h_{B-1}[L] & \cdots & h_{B-1}[0] \end{bmatrix}. \quad (12)$$

The i -th column vector of $\mathbf{E}_U$, $\mathbf{e}(\omega_i) = [e^{j\omega_i l_1}, e^{j\omega_i l_2}, \dots, e^{j\omega_i l_U}]^T$, resides in the signal subspace, thus it must be orthogonal to all the eigenvectors spanning the noise subspace. That is, $\mathbf{e}^H(\omega_i) \mathbf{v}_u = 0$, where $\mathbf{v}_u$ is the u -th eigenvector of $\mathbf{R}_l$, for $u = P+1, \dots, U$. Thus the normalized Doppler shifts can be estimated by identifying ω that can maximize the following MUSIC objective function,

$$J(\omega) = \frac{1}{\mathbf{e}^H(\omega) \mathbf{V}_l \mathbf{V}_l^H \mathbf{e}(\omega)}, \quad (20)$$

for $\omega \in \Omega = 2\pi T_s \cdot [-\nu_{\max}, \nu_{\max}]$, with $\nu_{\max}$ being the maximum Doppler spread and $\mathbf{V}_l = [\mathbf{v}_{P+1}, \dots, \mathbf{v}_U] \in \mathcal{C}^{U \times (U-P)}$ containing the last $U - P$ eigenvectors of $\mathbf{R}_l$.

C. Covariance Matrix Estimation

The estimation of the covariance matrix $\mathbf{R}_l$ is discussed in this subsection, and the results are used to develop a practical MUSIC algorithm for Doppler shift (s?) estimation.

From (19), the covariance matrix $\mathbf{R}_l$ depends on the Doppler shift ν_i , delay τ_i , and the power of the fading coefficients σ_i^2 , for $i = 1, \dots, P$. These channel parameters all change much slower compared to the fast time-varying fading coefficients. It is assumed that these parameters remain almost constant on the order of tens of milliseconds, which corresponds to around a dozen OTFS frames. Thus $\mathbf{R}_l$ can be estimated by using a running average over F consecutive OTFS frames, where F can be chosen to be around 10. At the n -th OTFS frame, the covariance matrix can be estimated as

$$\hat{\mathbf{R}}_l^{(n)} = \frac{1}{F} \sum_{i=n-F+1}^{i=n} \zeta_l^{(i)} \left(\zeta_l^{(i)} \right)^H \in \mathcal{C}^{U \times U}, \quad (21)$$

where $\zeta_l^{(i)}$ is the observed pilot vector from the i -th frame.

The accuracy of the estimation increases with the number of observations F . To further improve the estimation accuracy, we propose to decimate the observed pilot vector into D sub-vectors as

$$\zeta_{l,d} = [z_d[l], z_{d+D}[l], \dots, z_{d+(Q-1)D}[l]]^T \in \mathcal{C}^Q, \quad \text{for } d = 1, \dots, D, \quad (22)$$

where D is the decimation factor and it is chosen such that $Q = \frac{U}{D}$ is an integer, and we have $Q > P$ to ensure the stability of the MUSIC algorithm. The decimated pilot vector $\zeta_{l,d}$ is obtained by taking one out of every D samples of ζ_l .

From (14), we can see that the length- Q decimated pilot vector $\zeta_{l,d}$ samples the l -th diagonal of the channel matrix $\mathbf{H}$ with a sampling period $\frac{B}{U}DT_s = \frac{B}{Q}T_s$, over a duration of BT_s . For a given l , the decimated pilot vectors all share the same covariance matrix because

$$\mathbb{E}[h_u[l] h_v^*[l]] = \sum_{i=1}^P e^{j(u-v)\omega_i} \sigma_i^2 |A_q(lT_s - \tau_i, \nu_i)|^2 \quad (23)$$

depends only on the difference $(u - v)$.

The decimation of the pilot vector increases the number of vector samples by a factor of D . The covariance matrix of the decimated pilot vector can be estimated as

$$\hat{\mathbf{R}}_{lD}^{(n)} = \frac{1}{DF} \sum_{i=n-F+1}^{i=n} \sum_{d=0}^{D-1} \zeta_{l,d}^{(i)} \left(\zeta_{l,d}^{(i)} \right)^H \in \mathcal{C}^{Q \times Q}. \quad (24)$$

The covariance matrix $\hat{\mathbf{R}}_{lD}^{(n)}$ has a size $Q \times Q$ with $Q > P$. In this case, the signal vector residing in the signal subspace of $\hat{\mathbf{R}}_{lD}^{(n)}$ can be expressed as

$$\mathbf{e}_D(\omega) = [e^{j\omega l_1}, e^{j\omega l_1+D}, \dots, e^{j\omega l_1+(Q-1)D}]^T \in \mathcal{C}^Q, \quad (25)$$

which is obtained by decimating the original signal vector $\mathbf{e}(\omega)$ by a factor of D .

The MUSIC algorithm with the estimated covariance matrix $\hat{\mathbf{R}}_{lD}^{(n)}$ can then be described as

$$\hat{\nu}_i = \frac{1}{2\pi T_s} \arg \max_{\omega \in \Omega} \frac{1}{\mathbf{e}_D^H(\omega) \mathbf{V}_{Dl} \mathbf{V}_{Dl}^H \mathbf{e}_D(\omega)} \quad (26)$$

for $i = 1, \dots, P$, where $\mathbf{V}_{Dl} \in \mathcal{C}^{Q \times (Q-P)}$ contains the last $Q - P$ eigenvectors of $\hat{\mathbf{R}}_{lD}^{(n)}$.

D. Reconstruction of Channel Transfer Function

The equivalent discrete-time channel matrix $\mathbf{H}$ can be reconstructed by using the results of Doppler estimation and the observed pilot symbols. The fast time-varying time domain channel coefficients $h_k[l]$ can be decomposed as non-linear functions of the slow-changing DD domain channel parameters as in (9) and (15). Thus we propose to reconstruct $\mathbf{H}$ by exploiting the unique DD-domain structure of the channel coefficients.

We will first perform estimation of the coefficient vector β_l defined in (16), which incorporates the non-linear effects of the DD-domain channel parameters $\{\varphi_i, \nu_i, \tau_i\}_{i=1}^P$ and the ambiguity function $A_q(t, f)$. Based on the results of Doppler

estimation and (17), the coefficient vector β_l can be estimated by following the least squares (LS) criterion to minimize the following cost function

$$J(\beta_l) = \|\zeta_l - s_p \cdot \hat{\mathbf{E}}_U \cdot \beta_l\|_2^2 \quad (27)$$

where $\|\mathbf{a}\|_2^2 = \mathbf{a}^H \mathbf{a}$, and $\hat{\mathbf{E}}_U$ is formulated by replacing ω_i with the estimated Doppler shifts $2\pi\hat{\nu}_i T_s$ in (18). The solution is

$$\hat{\beta}_l = \frac{1}{s_p} \left(\hat{\mathbf{E}}_U^H \hat{\mathbf{E}}_U \right)^{-1} \hat{\mathbf{E}}_U^H \zeta_l. \quad (28)$$

Based on (15), the equivalent discrete-time channel coefficient can then be reconstructed as

$$\hat{h}_{k+l}[l] = [e^{j\hat{\omega}_1 k}, e^{j\hat{\omega}_2 k}, \dots, e^{j\hat{\omega}_P k}] \cdot \hat{\beta}_l. \quad (29)$$

All non-zero elements in the time domain channel matrix $\mathbf{H}$ can be reconstructed by using (29). The DD-domain channel transfer function can then be estimated by transforming $\mathbf{H}$ into the DD domain.

The estimation of both the Doppler shifts and the coefficient vector β_l requires knowledge of the number of paths P , which is usually not readily available at the receiver. We propose to tackle this problem by identifying the value of P that can minimize the difference between the estimated and observed channel coefficient at pilot locations. For any $P \in [1, P_{\max}]$, where $P_{\max}$ is the maximum possible number of paths for the physical channel, we can estimate the Doppler shifts $\{\nu_i\}_{i=1}^P$, the coefficient vector β_l , and the channel CIR $h_k[l]$. Then we can select P by minimizing the following cost function,

$$\hat{P} = \underset{P \in [1, P_{\max}]}{\operatorname{argmin}} \sum_{u=1}^U \sum_{l=0}^L \|z_u[l] - s_p \cdot \hat{h}_{l_u+l}[l]\|^2. \quad (30)$$

IV. SIMULATION RESULTS

The performances of OTFS systems with the proposed channel estimation algorithm are evaluated in this section. All systems employ QPSK modulation with rectangular prototype pulses. The carrier frequency is 4 GHz and the subcarrier spacing is set to $\Delta f = 15$ kHz. The physical channel is generated with an EVA delay profile [15] under a user equipment (UE) speed of 500 km/hr. The number of subcarriers and time symbols is fixed at $M = 64$ and $N = 32$, respectively. The detection is performed by using the successive interference cancellation minimum mean squared error (SIC-MMSE) receiver [16]. The maximum number of iterations of the SIC-MMSE receiver is set to 8, and most detections converge within 3 iterations.

In all simulations, the E_b/N_0 values are calculated by considering the energy overhead of the pilot symbols, that is, $E_b = \frac{E_s N M + E_p U}{N M \log_2 |\mathcal{M}|}$, where $|\mathcal{M}|$ is the cardinality of the modulation constellation set $\mathcal{M}$, and we have $E_s = E_p$. The throughput is calculated as

$$\eta_b = \frac{N M \log_2 |\mathcal{M}|}{N M + U(2L + 1)} (1 - \text{FER}), \quad (31)$$

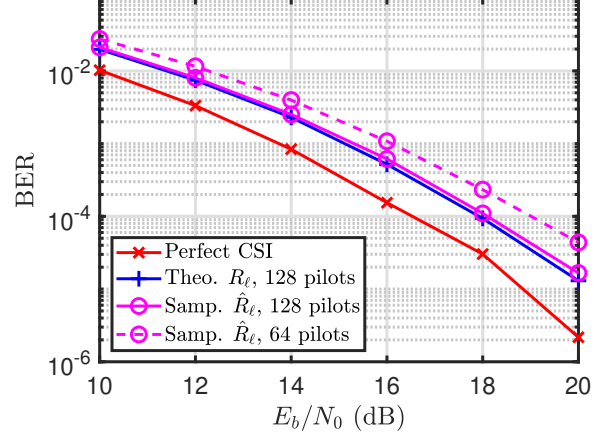


Fig. 2: BER with perfect and estimated CSI.

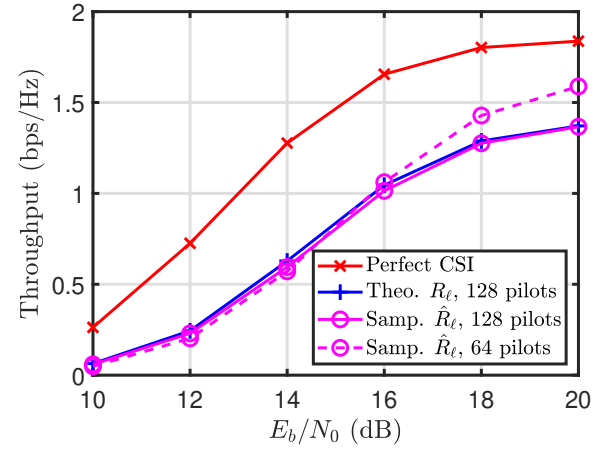


Fig. 3: Throughput with perfect and estimated CSI.

where FER is the frame error rate.

Figs. 2 and 3 compare the BER and throughput, respectively, for OTFS systems under various system configurations, including perfect CSI, channel estimations with the theoretical covariance matrix $\mathbf{R}_l$ (with 128 pilots), and channel estimations with estimated $\hat{\mathbf{R}}_l$. For systems with estimated $\hat{\mathbf{R}}_l$, 10 all-pilot frames were used for initialization of $\hat{\mathbf{R}}_l$, and the covariance matrix is then estimated by using the running average as shown in (24). Pilot decimation with $Q = 16$ is used for both $U = 64$ and 128. Each trial consists of 20 OTFS frames, and the results are averaged over 10,000 trials.

In the BER results in Fig. 2, when $U = 128$, the performance of systems with estimated $\hat{\mathbf{R}}_l$ is almost identical to that with ideal knowledge of $\mathbf{R}_l$, and this indicates that the proposed pilot decimation method can obtain an accurate estimate of $\mathbf{R}_l$. In addition, the performance of systems with 128 and 64 pilots are 1.5 and 2.1 dB away from that of systems with perfect CSI, respectively. In Fig. 3, the throughput of the system with 64 pilots is slightly better than that with 128 pilots due to lower pilot overhead.

The BER and throughput of various OTFS systems are shown as functions of the number of pilots per frame in Figs.

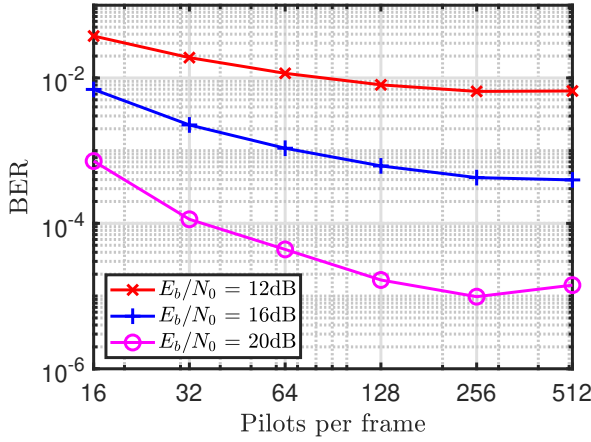


Fig. 4: Impacts of number of pilots on BER.

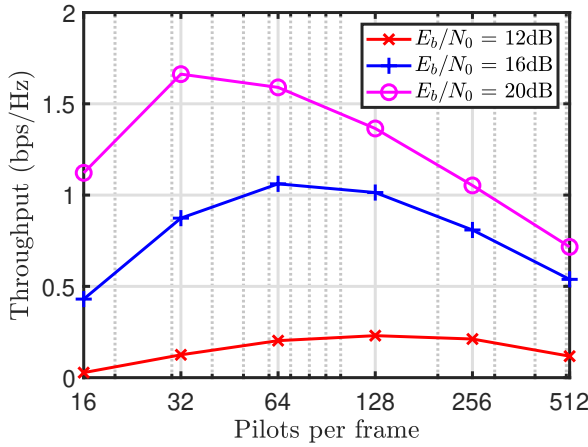


Fig. 5: Impacts of number of pilots on Throughput.

4 and 5, respectively. Pilot decimation with $Q = 16$ is used for all configurations. The BER performance is convex in the number of pilots U , with the optimum BER achieved at $U = 256$. When $U < 256$, the performance gain due to better channel estimation dominates the slight performance loss due to pilot energy overhead. This trend is reversed when U is too large. For example, with $M = 64$ and $N = 32$, increasing U from 256 to 512 incurs an effective E_b/N_0 loss of 0.46 dB due to pilot energy overhead.

The throughput in Fig. 5 is a concave function in the number of pilots U . The concave shape is caused by the tradeoff between channel estimation error and pilot overhead, as more pilots will reduce the system spectral efficiency. The maximum throughput is achieved at 128, 64, and 32 pilots for E_b/N_0 at 12 dB, 16 dB, and 20 dB, respectively. Fewer pilots are needed at higher E_b/N_0 as there is less noise distortion during the channel estimation process.

V. CONCLUSIONS

A new channel estimation algorithm with time-domain pilots has been proposed for OTFS systems. Unlike most previous works relying on DD-domain pilots with 2D guard intervals,

the proposed algorithm employed time-domain pilots with a 1D guard interval. The channel estimation algorithm was developed by using the MUSIC algorithm to exploit the unique non-linear relationship among time-domain channel coefficients and DD-domain propagation parameters. The tradeoff between pilot overhead and system performance was evaluated by using simulations, which have demonstrated that the OTFS system with the proposed channel estimation algorithm can achieve a performance that is close to that with perfect CSI.

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